

Math 4441: Probability and Statistics

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Chapter 1: Introduction (Part 1 – Probability)

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Course Outcomes (Cos)

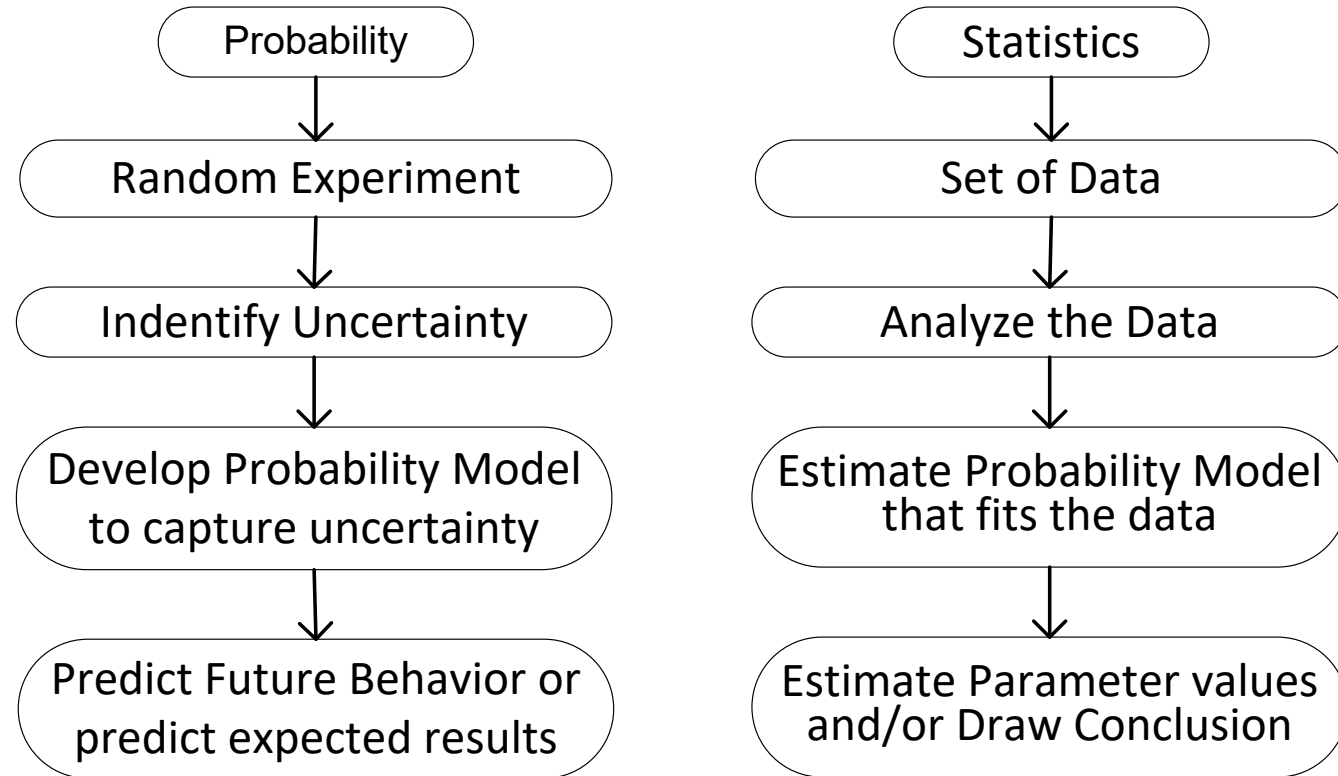
- CO1** Understand fundamental knowledge of probability, including probability of events, conditional probability, sum and product rules, Bayes rule, random variables, expectation and variance, and correlation and covariance.
- CO2** Apply knowledge of probability to model solutions for complex engineering problems or part of them and mathematically explain uncertain phenomenon.
- CO3** Use appropriate probability models to describe and analyze observed data, and draw conclusions concerning interesting parameters

Chapter 1: Introduction

Contents:

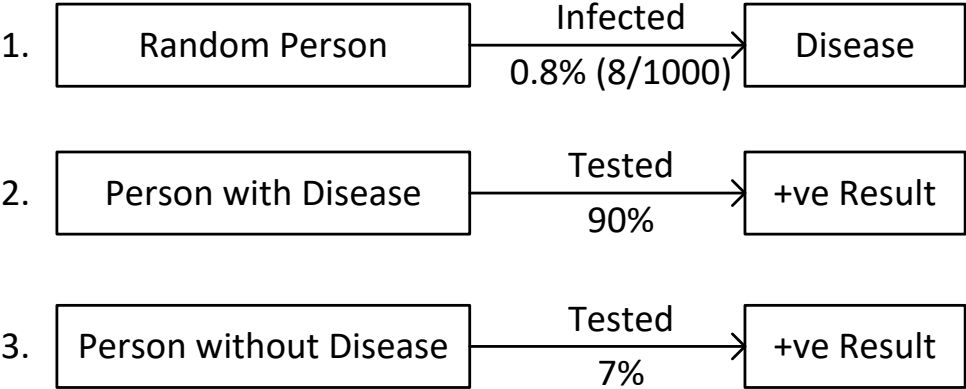
- Part 1 – Probability
 - Basic concepts of probability
 - Simple probability model
 - Joint probability
- Part 2 – Conditional Probability
 - Concept of conditional probability
 - Product and sum rules of probability
 - Bayes Theorem
 - Independence of events

Probability vs Statistics



Problem 1 (Disease or Not Disease)

The probability that a randomly selected person has a disease is 0.8 percent (8 out of 1,000). The probability that a person with the disease will have a positive test result is 90 percent. The probability that a person without the disease will have a positive test result is 7 percent. Find the probability that a person is infected given that he is tested positive.

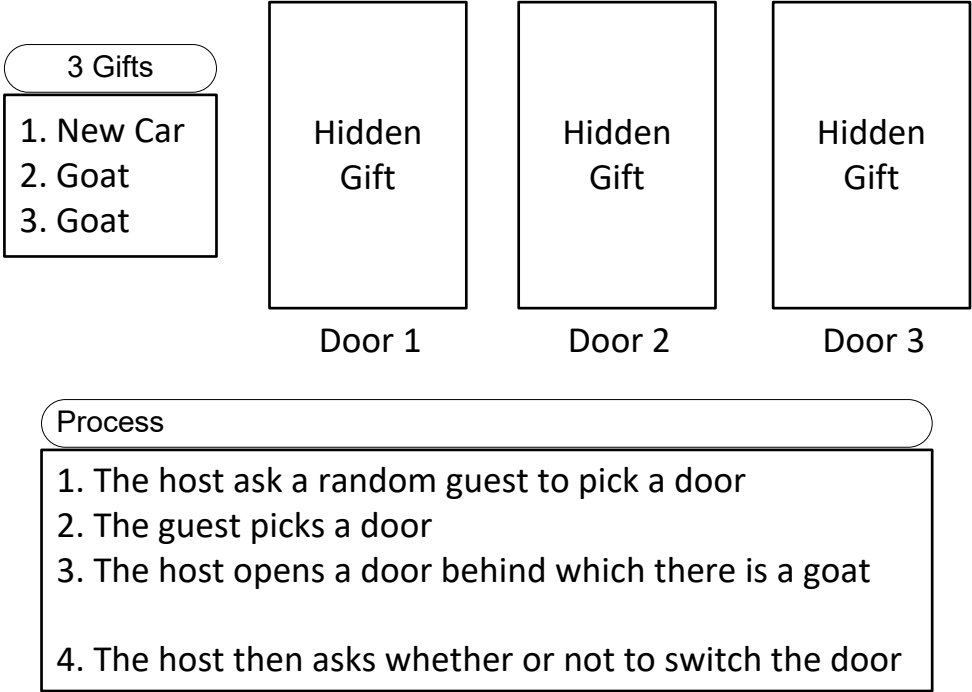


Problem 2 (Monty Hall Dilemma)

The dilemma is taken from an old American game show called Let's Make a Deal. The original host of the show, Monty Hall, would select a member of the audience and show that person three large closed doors labeled 1, 2, and 3. Behind one of the doors was a new car. Behind the remaining two doors were joke prizes, such as a live goat.

The contestant was asked to pick a door. Then, Hall would ask that one of the doors the contestant didn't pick be opened, naturally one that didn't have a car behind it. After the audience stopped laughing at whatever joke prize was behind that door, Hall would ask the contestant if they wanted to keep the originally selected door, or if they would rather change their selection to the remaining door. The dilemma is simply that: do they keep their original guess, or do they switch to the remaining door?

Switch or Not to Switch



Problem 3 (Birthdays Problem)

Determine the probability that in a class of size r there are two or more students with the same birthday. Assume that the year consists of 365 days.

Probability and Probability Models

Probability

- A probability is a number between 0 and 1, and represents the likelihood of occurring some uncertain phenomenon.
- A probability quantifies the chances by a number between 0 and 1.

Probabilistic Models or Probability Models

- A probability model is a mathematical description of an uncertain situation.
- Quantifies the likelihood of some uncertain phenomenon mathematically.

Certainty vs Uncertainty (Number of possible distinct results)

Random Experiments

Associated with multiple possible results and produce exactly one result randomly when the experiment is run.

Experiments consist of two ingredients:

Procedure: Real world phenomenon that produces uncertain outcomes.

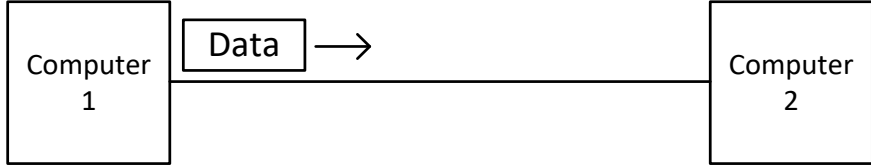
Observation: Uncertain results that we are interested in

Each of the distinct results of an experiment is an **outcome**.

A set of outcomes is an **event**

If any outcome associated with an event occurs, we say the event has occurred

Example 1.1 Sending a data packet from one computer (sender) to another computer (receiver). Success occurs if the packet is Delivered (D) and failure occurs if the packet is not delivered (F)



Example 1.1

Procedure: Send the data packet from the sender to the receiver

Observation: Observe the success or failure of the delivery.

Probability Model: Likelihood of **D** is three times than that of **F**. Each of the deliveries is independent of the others → One does not affect others

Similarity of Experiments

Experiment	Procedure	Observation
Example 1.2	Send 3 packets from one computer to another	Sequence of success (D) and failure (F)
Example 1.3		Number of successes

Experiment	Procedure	Observation
Example 1.4	Keep sending packets from one computer to another until the first successful delivery	Number of packets sent
Example 1.5	Keep sending packets from one computer to another until 3 packets are delivered	

Two experiments are

- similar *iif* both procedures and observations are similar.
- different *if* either or both the procedures and/or the observations are different.

Same procedure but different observations

Experiment 1.2

Procedure: Send 3 packets from a sender to a receiver.

Observation: Sequence of successes and failures of the deliveries.

Experiment 1.3

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

Same observation but different procedures

Experiment 1.4

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

Experiment 1.5

Procedure: Keep sending packets from a sender to a receiver until 3 packets are delivered

Observation: Number of attempts

Outcomes, Events and Probability

Outcome: Every possible result of an experiment

An outcome is an observation

Event: A set of one or more outcomes

Probability: Percentage of time an outcome of event occurs

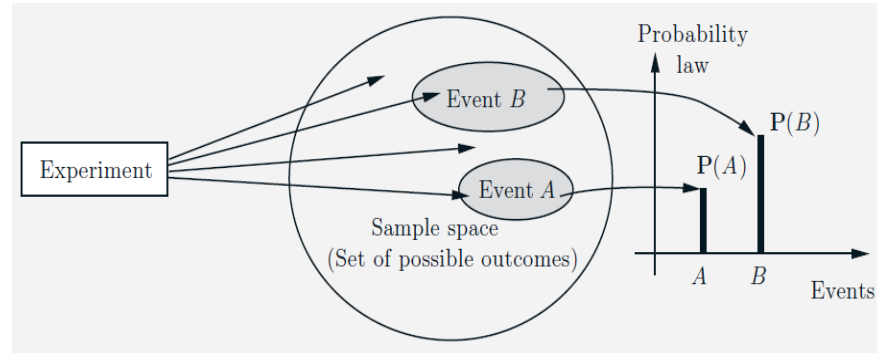
$P[A] \rightarrow$ probability of occurring the event A

$P[\omega] \rightarrow$ Probability of occurring the outcome ω

Elements of Probability Model

→ Sample Space (S)

→ Probability of outcomes/events ($P[\cdot]$)



Elements of Probability Model

→ Sample Space (S): set of all possible outcomes

→ Probability of outcomes/events: Assignment of probabilities to each outcome of event

Sample Space A set of elements is said to be the *sample space*

→ if the elements individually/collectively satisfy 3 properties

1. Finest Grain

- An outcome must be explicit and refined
- Any result of the experiment points to a single element of S
- If $\omega \in S$, any subset of ω cannot be an element of S

2. Mutually Exclusive

- Elements in S are disjoint, $\omega_i \cap \omega_j = \phi$
- If ω is an outcome, then none of $S \setminus \omega$ can be an outcome at the same time

3. Collectively Exhaustive

- Every outcome of the experiment must be in S
- Probability of occurring an outcome from S must be 1

Event Space

Collectively exhaustive and mutually exclusive set of events

$E = \{E_1, E_2, \dots, E_n\}$ is an event space, if

- Mutually exclusive: All events in E are disjoint

$$E_i \cap E_j = \phi, \text{ if } i \neq j$$

- Collectively exhaustive: Outcomes in E must include all elements in S

$$\bigcup_{i=1}^n E_i = S$$

Sample Space

- 1. Discrete Set
- 2. Continuous Set

Example of Sample Space

Experiment	Procedure	Observation
1.2	Send 3 packets from one computer to another	Sequence of success (D) and failure (F)
1.3		Number of successes

Experiment 1.2

$$S = \{FFF,FFD,FDF,FDD,DFF,DFD,DDF,DDD\}$$

Experiment 1.3

$$S = \{FFF,FFD,FDF,FDD,DFF,DFD,DDF,DDD\}$$

Experiment 1.6: Example of Event Space

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

$$E_i = \{i \text{ successes}\}, \quad i = 0, 1, 2, 3$$

$$E_0 = \{0 \text{ success}\} = \{FFF\}$$

$$E_1 = \{1 \text{ success}\} = \{FFD, FDF, DFF\}$$

$$E_2 = \{2 \text{ successes}\} = \{FDD, DFD, DDF\}$$

$$E_3 = \{3 \text{ successes}\} = \{DDD\}$$

$$E = \{E_0, E_1, E_2, E_3\} \text{ is an event space or not?}$$

Example of Sample Space

Experiment	Procedure	Observation
1.4	Keep sending packets from one computer to another until the first successful delivery	Number of packets sent
1.5	Keep sending packets from one computer to another until 3 packets are delivered	

Experiment 1.4

$$S = \{D, FD, FFD, FFFD, \dots\}$$

Experiment 1.5

$$S = \{DDD, FDDD, DFDD, DDFD, FFDDD, \dots\}$$

Example 1.7: Monitor 3 consecutive phone calls going through a telephone switching office. Classify each one as a voice call (v) if someone is speaking, or a data call (d) if the call is carrying a modem or fax signal. Your observation is a sequence of three letters (each letter is either v or d). For example, two voice calls followed by one data call corresponds to vvd .

Write the elements of the following sets:

1. $A_1 = \{\text{first call is a voice call}\}$
2. $B_1 = \{\text{first call is a data call}\}$
3. $A_2 = \{\text{second call is a voice call}\}$
4. $B_2 = \{\text{second call is a data call}\}$

$$S = \{vvv, vvd, vdv, vdd, dvv, dvd, ddv, ddd\}$$

$$S = \{vvv, vvd, vdv, vdd, dvv, dvd, ddv, ddd\}$$

$$A_1 = \{vvv, vvd, vdv, vdd\}$$

$$B_1 = \{dvv, dvd, ddv, ddd\}$$

$$A_2 = \{vvv, vvd, dvv, dvd\}$$

$$B_2 = \{v dv, v dd, ddv, ddd\}$$

5. $A_3 = \{\text{all calls are the same}\}$

$$A_3 = \{vvv, ddd\}$$

6. $B_3 = \{\text{voice and data alternate}\}$

$$B_3 = \{vdv, dvd\}$$

7. $A_4 = \{\text{one or more voice calls}\}$

$$A_4 = \{vvv, vvd, vdv, vdd, dvv, dvd, ddv\}$$

8. $B_4 = \{\text{two or more data calls}\}$

$$B_4 = \{vdd, dvd, ddv, ddd\}$$

Probabilities of Events/Outcomes:

The probability of occurring event A is defined by the proportion of S represented by A

➔ How probabilities are assigned to the outcomes and events

1. Probability Axioms

- a. Rules related to the assignment of probabilities, and every assignment must satisfy
- b. Ensures a number between 0 and 1 to each outcome and event
- c. Ensures with probability one of the outcomes of S occurs

2. Assignment of Probabilities

- a. Classical Approach
- b. Relative Frequency Approach
- c. Subjective Approach

Probability Axioms: Rules that every prob. assignment must satisfy

Probability Axioms: Rules that every prob. assignment must satisfy

Axiom 1 (Non-negativity): If A is an event, then the probability of A satisfies

$$P[A] \geq 0$$

Axiom 2 (Additivity): If A_1 and A_2 are two disjoint events, then

$$P[A_1 \cup A_2] = P[A_1] + P[A_2]$$

Axiom 3 (Normalization): The probability of all elements in the sample space must be sum to 1

$$P[S] = P[\cup_{i=1}^n \omega_i] = \sum_{i=1}^n P[\omega_i] = 1$$

Simple Extension of the Axioms

1. If $A_1, A_2, \dots, A_n$ is a disjoint set, then

$$P[\cup_{i=1}^n A_i] = P[A_1] + P[A_2] + \dots + P[A_n]$$

2. If $A \cap B \neq \phi$, A and B are not mutually exclusive, then

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

3. $P[A^c] = 1 - P[A]$

4. If $A \subset B$, then $P[A] \leq P[B]$

Assignment of Probabilities

1. Classical Approach

- a. Probability of an outcome
- b. Probability of an event

The classical approach assumes that all outcomes are equally likely

- If the coin is fair, heads and tails are equally likely to appear
- If the die is fair, number of dots 1, 2, ..., 6 are equally likely to appear

Probability of Outcomes: Let there are n elements in the sample space

$$S = \{\omega_1, \omega_2, \dots, \omega_n\}$$

Axiom 1: Non-negative

$$P[\omega_i] \geq 0 \text{ [axiom 1]}$$

Axiom 2: ω_i s are elements of S , $\omega_i \cap \omega_j = \phi$, if $i \neq j$

$$P[\omega_1 \cup \omega_2 \cup \dots \cup \omega_n] = P[\omega_1] + P[\omega_2] + \dots + P[\omega_n]$$

Axiom 3: $P[S] = 1$

$$P[\omega_1] + P[\omega_2] + \dots + P[\omega_n] = 1$$

Since the outcomes are equally likely

$$P[\omega_1] = P[\omega_2] = \dots = P[\omega_n]$$

$$P[\omega_i] = \frac{1}{n}, i = 1, 2, \dots, n$$

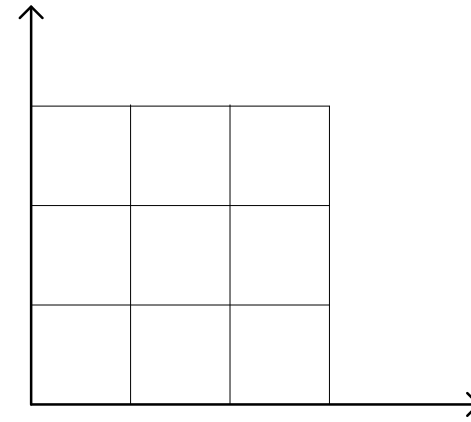
Probability of Events: Classical Approach

The probability of occurring event A is,

1. If the outcomes are equally likely, the ratio of the number of outcomes in A to the number of outcomes in S
2. If the outcomes are not equally likely?

Example 1.8 A pair of four-sided fair dice are rolled and the number of dots on the side facing upward in each dice is counted. Each outcome of this experiment is a pair of numbers in the range $1, 2, \dots, 4$. There are 16 possible outcomes of this experiment. If we assume, if the outcomes are equally likely to occur, then each outcome is assigned a probability of $\frac{1}{16}$.

Find the probability of the event $A = \{\text{sum of two dice} = 5\}$



Probabilities of some events:

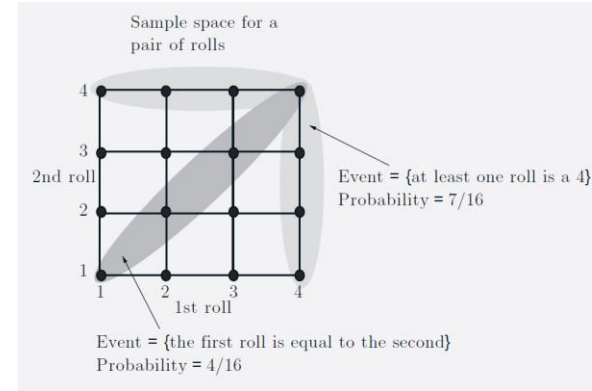
1. $P[\text{the sum of the rolls is even}] =$

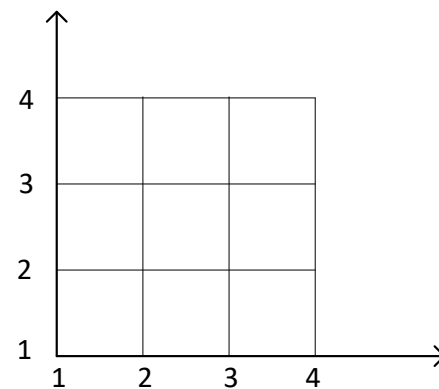
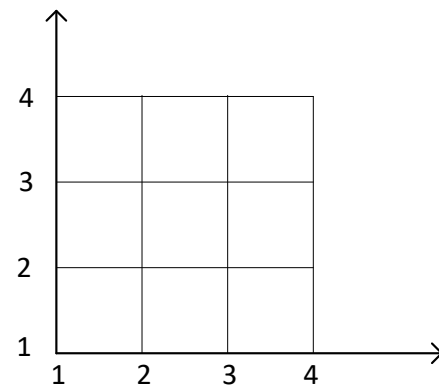
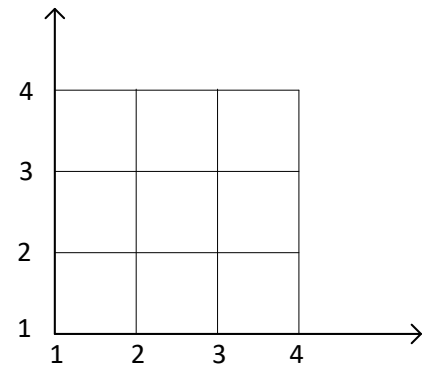
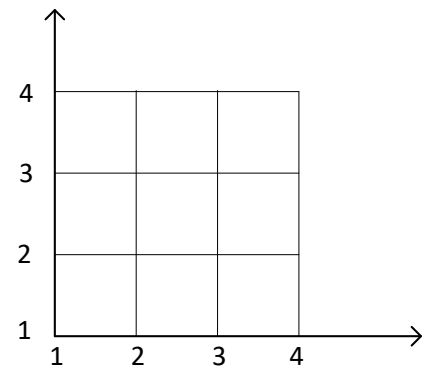
2. $P[\text{the sum of the rolls is odd}] =$

3. $P[\text{the first roll is equal to the second}] =$

4. $P[\text{the first roll is larger than the second}] =$

5. $P[\text{at least one roll is equal to 4}] =$





2. Relative Frequency Approach

- a. If the outcomes in the sample space are not equally likely
- b. We repeat the experiment a large number of times (say, n and $n \rightarrow \infty$)
- c. Count the number of times outcome ω occurs, n_ω

The probability of occurring outcome ω is

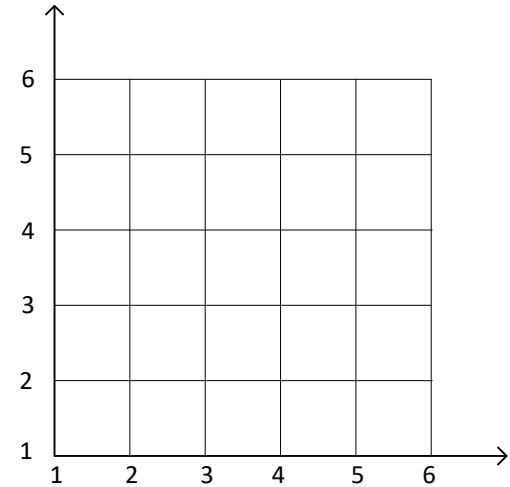
$$P[\omega] = \lim_{n \rightarrow \infty} \frac{n_\omega}{n}$$

Similarly, the probability of occurring any event A can be calculated by counting the number of times event A occurs n_A if the experiment is repeated n times,

$$P[A] = \lim_{n \rightarrow \infty} \frac{n_A}{n}$$

Example 1.9: Consider two six-sided fair dice are rolled. We define event $E_k = \{\text{sum of the two dice} = k\}$. We can simulate the experiment by a computer program, and the results are shown below for the event $A = E_5$:

n	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
n_A	96	200	314	408	521	630	751	859	970	1095
n_A/n	0.096	0.10	0.105	0.102	0.104	0.105	0.107	0.107	0.108	0.110



$$\begin{aligned}
 P[A] &= P[\{1,4\} \cup \{2,3\} \cup \{3,2\} \cup \{4,1\}] \\
 &= P[\{1,4\}] + P[\{2,3\}] + P[\{3,2\}] + P[\{4,1\}] \\
 &= \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{4}{36} = \frac{1}{9} = 0.111
 \end{aligned}$$

3. Subjective Approach

- a. Probabilities are assigned by engaging subject experts
- b. Manually assign the probabilities
- c. Vary from expert to expert.

Joint Probability: (Prob. of the simultaneous occurrence of 2 or more events)

The probability of the intersection of two events, $P[A \cap B]$ or $P[A, B]$ or $P[AB]$

It is the simultaneous occurrence of two events

If events A and B are mutually exclusive, they two cannot occur together and the probability $P[AB] = 0$, because $A \cap B = \phi$

Calculation of the joint Probability:

1. Classical Approach: Find $A \cap B = \{\text{elements common in both events}\}$

$$P[A \cap B] = P[\cup_{\omega_i \in A \cap B} \omega_i] = \sum_{\omega_i \in A \cap B} P[\omega_i]$$

2. Use the relative frequency approach. If $n_{A,B}$ is the number of times events A and B occur simultaneously in n repetition, then

$$P[A, B] = \lim_{n \rightarrow \infty} \frac{n_{A,B}}{n}$$

Example 1.10: A standard deck of playing cards has 52 cards that can be divided in several manners. There are four suits (spades, hearts, diamonds, and clubs) each of which has 13 cards (Ace, 2, 3, ... , 10, jack, queen, and king). There are two red suits (hearts and diamonds) and two black suits (spades and clubs). Also, the jacks, queens, and kings are referred to as face cards, while the others are number cards.

1. $A = \{\text{red card selected}\}$,
2. $B = \{\text{number card selected}\}$, and
3. $C = \{\text{heart selected}\}$.

Example 1.11: Roll a six-sided fair die and observe the number of dots on the top side

$$S = \{1, 2, 3, 4, 5, 6\}$$

Define two events:

$$A \triangleq \{\text{Outcome is square of an integer}\} = \{1, 4\}$$

$$B \triangleq \{\text{Outcome is an even number}\} = \{2, 4, 6\}$$

$$P[A] =$$

$$P[B] =$$

$$P[AB] =$$

Simple Probability Models: Example 1.11 (Experiment 1.2)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Sequence of successes and failures of the deliveries.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

Probability that a packet is delivered is $\frac{1}{2}$

Probability that a delivery is failed is $\frac{1}{2}$

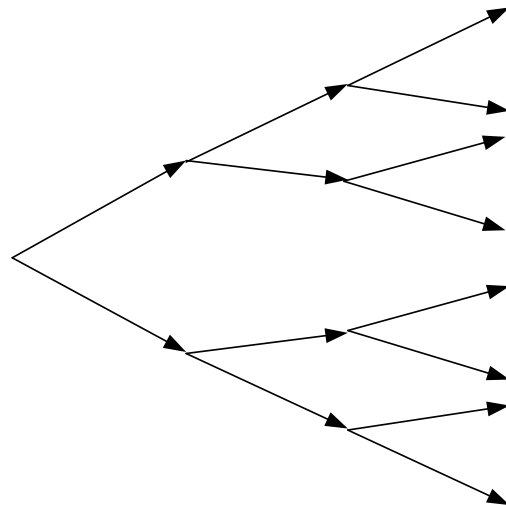
Probability of occurring a sequence of events

$$P[FFF] =$$

Probability that a packet is delivered is p

Probability that a delivery is failed is $1 - p$

$P[FFF] =$

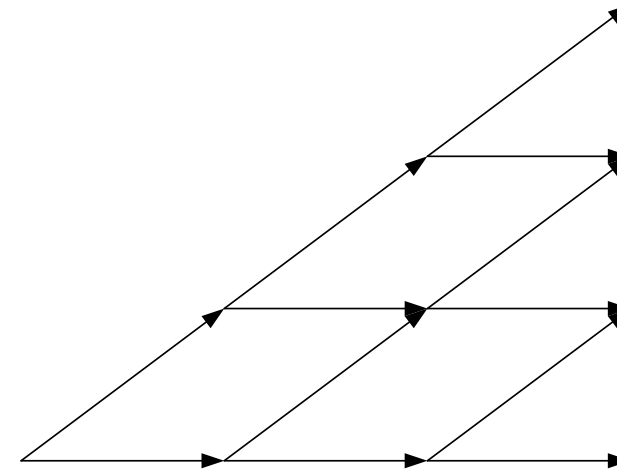


Example 1.12 (**Experiment 1.3**)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

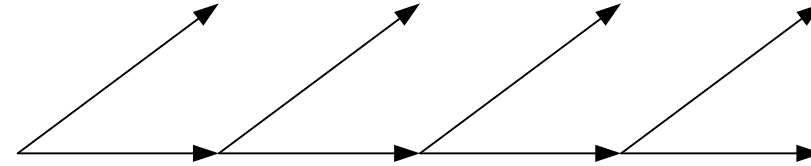


Example 1.13 (**Experiment 1.4**)

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

$$S = \{D, FD, FFD, FFFD, \dots\}$$



Example 1.14 (**Experiment 1.5**)

Procedure: Keep sending packets from a sender to a receiver until 3 packets are delivered

Observation: Number of attempts

$$S = \{DDD, FDDD, DFDD, DDFD, FFDDD, \dots\}$$

